

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

**ENVIRONMENTAL SCIENCE****Paper 2 (Generated practice paper - Set 37)**

Assessed: The living environment, Biological resources, Sustainability, Research methods

**Time allowed: 3 hours****Materials**

For this paper you may use:

- a calculator.

**Instructions**

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

**Information**

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

**For Examiner's Use**

| Question | Mark |
|----------|------|
| 1        |      |
| 2        |      |
| 3        |      |
| 4        |      |
| 5        |      |
| 6        |      |
| 7        |      |
| 8        |      |
| 9        |      |
| 10       |      |
| 11       |      |
| TOTAL    |      |

**Answer all questions in the spaces provided.**

**01 . 1**

Explain how the relationship between coral polyps and zooxanthellae allows reefs to support a high biomass in nutrient-poor tropical seas.

**[4 marks]**

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**01 . 2**

Describe the abiotic conditions needed by reef-building corals and explain why these make reefs vulnerable.

**[3 marks]**

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**01 . 3**

Explain **three** ways in which coral reefs are important to humans.

**[3 marks]**

**1** .....

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**2** .....

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**3** .....

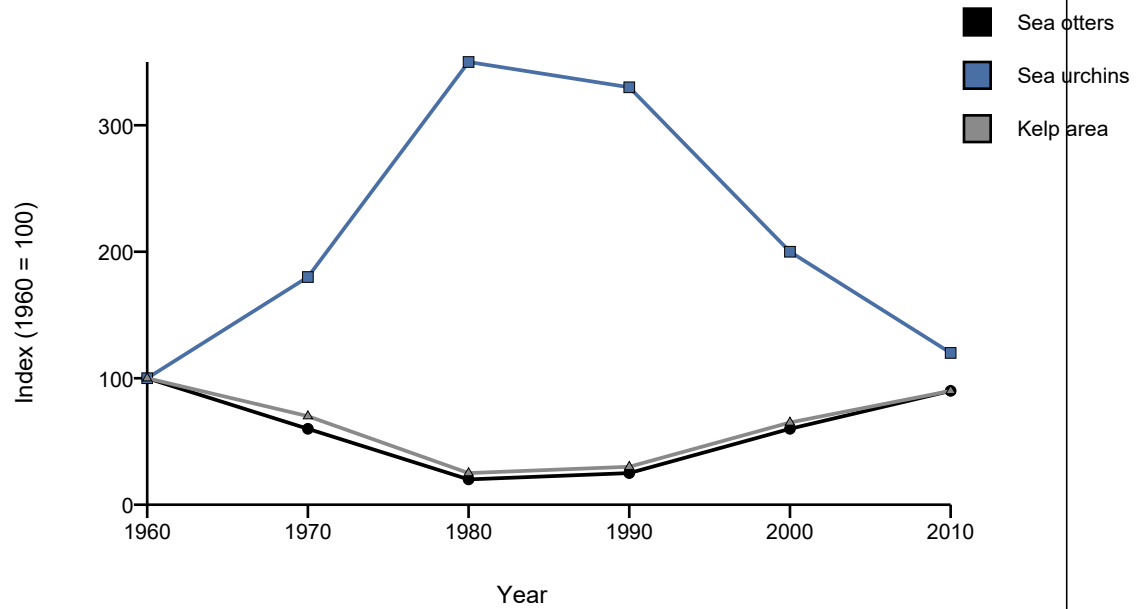
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10

0 2

Figure 1 shows the estimated populations of sea otters, sea urchins and the area of kelp forest along part of the coast of California between 1960 and 2010.

**Figure 1**



0 2

1 Describe the relationship between the three variables shown in Figure 1.

**[3 marks]**

0 2

2 Explain the relationships shown in Figure 1.

**[3 marks]**

0 2 . 3

Give **two** other examples of human over-exploitation of one species affecting the population of another.

[2 marks]

1

2

0 2 . 4

Explain why the loss of beavers or forest elephants can cause the decline of many other species.

[2 marks]

10

0 3 . 1

Explain the effects of deforestation on soil.

[4 marks]

0 3 . 2

Explain how deforestation affects hydrology and climate.

[4 marks]

03 . 3

Give **two** ecological impacts of deforestation other than the direct loss of species.

[2 marks]

1 .....

2 .....

10

04

Table 1 shows the mean dry mass of wheat plants grown with two fertiliser treatments (10 replicates each).

Table 1

| Treatment                | Mean dry mass / g | Standard deviation / g |
|--------------------------|-------------------|------------------------|
| Organic manure           | 14.2              | 3.9                    |
| Inorganic NPK fertiliser | 16.1              | 4.1                    |

04 . 1

Explain what the standard deviation values tell you about the data.

[2 marks]

.....

.....

.....

04 . 2

Name the statistical test that should be used to decide whether the difference between the two means is significant, and justify your choice.

[2 marks]

0 4 . 3

The test gave a probability  $p = 0.30$ . State what this means and what the student should conclude.

[2 marks]

0 4 . 4

Explain why a statistically significant correlation does not prove a causal relationship.

[2 marks]

0 4 . 5

Suggest how the student could increase the chance of detecting a real difference if one exists.

[2 marks]

10

**0 5**

Table 1 shows climate data for two sites where a farmer is considering growing maize, a frost-sensitive crop that needs a thermal growing season of at least 120 days.

**Table 1**

| Feature                               | Site A: valley floor | Site B: south-facing slope |
|---------------------------------------|----------------------|----------------------------|
| Frost-free period / days              | 110                  | 150                        |
| Days with mean temperature above 5 °C | 160                  | 175                        |
| Mean summer rainfall / mm             | 180                  | 150                        |
| Gradient / %                          | 1                    | 8                          |

**0 5****. 1**

Use Table 1 to explain which site is more suitable for maize.

**[3 marks]**

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**0 5****. 2**

Describe **three** ways in which temperature can be controlled to increase crop productivity.

**[3 marks]****1****2****3****0 5****. 3**

Explain why keeping livestock warm increases the amount of food converted to growth.

**[2 marks]**

.....

.....

0 5 . 4

Explain how day length affects livestock production and how it can be manipulated.

[2 marks]

10

0 6 . 1

Which of the following is a feature of natural systems? Tick **one** box.

[1 mark]

A Reactions that require high temperatures

☐

B Reliance on non-renewable energy

☐

C Waste products that are the raw materials for other processes

☐

D Dispersal of elements so they cannot be re-used

☐

0 6 . 2

Explain why nitrogen fixation by bacteria is more sustainable than the Haber process.

[2 marks]

0 6 . 3

Explain why incineration of wastes is less sustainable than natural decomposition.

[2 marks]

5

**0 7**

The Isle of Eigg (area 31 km<sup>2</sup>, population about 100) is not connected to the national grid. Table 1 shows its electricity sources.

**Table 1**

| Source                       | Capacity / kW |
|------------------------------|---------------|
| Hydroelectric (3 generators) | 110           |
| Solar PV                     | 50            |
| Wind (4 turbines)            | 24            |
| Diesel generators (back-up)  | 160           |

**0 7****1**

Explain why Eigg uses a mix of three renewable resources rather than one, and why diesel back-up is still needed.

**[3 marks]**

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**0 7****2**

Describe **three** other ways in which the Eigg community has increased its sustainability.

**[3 marks]****1**

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**2**

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**3**

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**0 7****3**

Explain why small self-contained communities often find it easier to adopt sustainable lifestyles than large cities.

**[2 marks]**

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07 . 4

Explain how large-scale DC electricity grids such as Desertec could increase the use of renewable energy.

[2 marks]

10

08 . 1

Explain how the development of biogeochemical cycles and of transpiration by land plants helped to maintain stable conditions for life.

[3 marks]

08 . 2

Describe **three** limitations of early methods of studying past conditions on Earth.

[3 marks]

1 .....

2 .....

3 .....

08 . 3

Describe how each of the following modern techniques provides information about past conditions.

[4 marks]

Isotope analysis of ice cores .....

Gas analysis of ice cores .....

Radiocarbon dating .....

Magnesium : calcium ratio in calcite .....

10

09 . 1

Explain the difference between in-situ and ex-situ conservation and state when ex-situ conservation is needed.

[2 marks]

09 . 2

Give **four** criteria used to select species for captive breeding and release programmes.

[4 marks]

1

2

3

4

09 . 3

Explain why each of the following species would be difficult to keep or breed in captivity.

[4 marks]

**Blue whale** .....

.....

**Large blue butterfly** .....

10

**1 0 . 1**

Explain how an understanding of ecological succession and population dynamics can inform conservation decision-making.

**[9 marks]**

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Extra space

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|---|---|
| 1 | 0 |
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| 2 |
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 State what is meant by a **taxon**.

**[1 mark]**

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**Write an essay on one of the following topics.**

|   |   |   |   |
|---|---|---|---|
| 1 | 1 | . | 1 |
|---|---|---|---|

Using tropical coral reefs as an example, discuss how the ecological features of a habitat determine both its importance and its vulnerability to human activities.

**[25 marks]**

**OR**

1 1 . 2

Discuss the causes and consequences of deforestation and evaluate the extent to which forests can be exploited sustainably.

**[25 marks]**

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.



This image shows a full page of primary-ruled paper. It features a series of horizontal dashed lines spaced evenly down the page, designed for handwriting practice. A single solid line runs horizontally across the top of the page, serving as a baseline or header area. The rest of the page is white, providing ample space for writing.



**END OF QUESTIONS**